JOHN K. RYU

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Full contact information available upon request

PROFESSIONAL EXPERIENCE

Forward Deployed Engineer | TecAce Software Ltd *Bellevue, WA | July 2026 - Present*

Embedded on site with a multinational direct-to-consumer manufacturer (US, Canada, Korea) to lead its AI transformation and modernize its ERP, e-commerce and logistics systems.

- **Lead customer discovery with executives and operations, logistics, finance, customer support and marketing teams**; mapped order-to-cash, purchasing and fulfillment workflows into a scoped, prioritized delivery roadmap
- **Co-designed and deliver a 12-week AI adoption program** covering Claude Team rollout and administration, on-site hands-on training in Claude, Claude Code and custom skills, and weekly bilingual (English/Korean) executive status reports
- **Built and deployed a hosted multi-store Shopify MCP server** on Google Cloud Run with Google OAuth sign-in and per-user Shopify staff permissions, giving staff governed AI read/write access across four storefronts; added multi-account Gmail and Google Drive MCP servers
- **Audited Shopify, Intercom, NetSuite and Celigo iPaaS integrations**; traced recurring sync failures to root causes, including concurrency set above NetSuite's governance limit, and delivered a prioritized remediation plan to leadership
- **Integrated NetSuite ERP through a least-privilege role and certificate-based OAuth 2.0 M2M access**; building the data flow between NetSuite and the Korea headquarters ERP (Ecount) and delivering API access to the engineering team
- **Built and shipped a bilingual executive operating dashboard** for the general manager (Next.js, TypeScript, PostgreSQL + pgvector, Google Cloud Run) that consolidates sales, orders, inventory and PSI planning from ERP and commerce systems, with an AI agent and daily briefs, plus a public demo on synthetic data
- **Architecting an internal operations platform on Google Cloud** (Cloud Run, Cloud SQL, Secret Manager) with Google SSO, RBAC, audit logging, CI/CD and human approval before any ERP write-back; first workloads automate proforma invoice validation, PO import and shipment data entry into NetSuite
- **Scoped carrier and 3PL integrations** (FedEx API/EDI, ShipStation API, a Canadian 3PL's WMS API); ran API gap analyses, wrote technical requirements and steered vendors toward a design with NetSuite as the source of truth
- **Proposed a company-wide ERP architecture** across the US, Canada, Korea and Europe, backed by a total-cost and valuation case presented to leadership

AI Engineer | TecAce Software Ltd *Bellevue, WA | April 2025 - June 2026*

- **Architected a governance-first, privacy-first enterprise AI agent platform** with a retrieval-augmented generation (RAG) pipeline (LangChain, LangSmith, LangGraph), deployable on hybrid infrastructure — cloud (AWS) or fully on-prem with self-hosted model serving — so regulated teams keep sensitive data inside their own perimeter
- **Engineered LangGraph-based agentic workflow platform** — visual flow builder with intelligent routing, guardrails, conversation memory, and node-level inference overrides powering multi-agent enterprise applications
- **Built messaging-platform integration gateway** connecting Slack and Microsoft Teams to AI applications with signed webhooks, multi-app alias routing, and real-time SSE streaming replies
- **Developed hybrid search optimization** implementing RRF reranking, cross-encoder reranking, and score fusion techniques for enhanced search relevance across Milvus vector databases
- **Built an end-to-end evals (LLM evaluation) system for RAG pipelines and AI agents** using multiple eval methods — synthetic test-set generation (RAGAS TestsetGenerator, DeepEval Synthesizer), versioned golden datasets, and an eval runner that replays test cases through the live RAG pipeline, capturing answers, retrieved documents, and citation scores; applied the same evals while developing AI agents
- **Combined algorithmic, model-graded, and regression evals** — retrieval metrics (Precision@k, Recall@k, MRR, NDCG, source coverage), a configurable LLM-as-judge layer (AWS Bedrock or on-prem vLLM) scoring faithfulness, answer relevance, and hallucination rate, benchmark validation against FinanceBench, A/B run comparison, known-bad-prompt regression suites, and automated drop-vs-baseline detection
- **Delivered external developer platform** — access-key-secured public API gateway plus a complete data-driven API reference documentation portal covering 500+ endpoints across six microservices
- **Championed AI-native engineering practice** orchestrating coding agents (Claude Code) through spec → plan → multi-agent implementation → adversarial QA workflows across a 12-repository platform

e-Commerce Engineer/Owner | Roem Ventures LLC *Seattle, WA | May 2022 - Present*

- **Founded and developed e-commerce brand from concept to market launch** including comprehensive product development, logistics coordination, and business operations management
- **Engineered custom Shopify solutions using Liquid programming** developing tailored plugins and functionality enhancements for optimized user experience
- **Implemented integrated marketing campaign strategies** utilizing Google AdWords and Facebook Advertising platforms for targeted customer acquisition and conversion optimization

Software Engineer - Xfinity X1 & Flex | Comcast *Philadelphia, PA | Oct 2017 - Nov 2021*

- **Successfully developed and maintained Xfinity Flex Streaming product from ground up** building key middleware APIs that power core Flex features, resulting in positive customer feedback
- **Exceeded subscription acquisition targets by 1000%** surpassing initial goal of 100K subscriptions within six months by acquiring over 1 million subscriptions
- **Developed and maintained core features for cloud-based X1 Guide interface** directly impacting over 31 million customers across Xfinity TV platform
- **Implemented scalable backend architecture** using distributed systems design principles to support millions of concurrent streaming users

Startup Founder | Parkgorithm *Philadelphia, PA | May 2016 - Jun 2017*

- **Founded startup tackling urban street parking through mobile technology** — led cross-functional team, product roadmap, and Android MVP delivery

Software Engineer | Graphite GTC *Bryn Mawr, PA | May 2015 - May 2016*

- **Developed Android and Web client applications** for the proprietary Graphite platform — responsive, cross-platform UIs with UX and performance optimization

Undergraduate Researcher | Temple University *Philadelphia, PA | Jun 2015 - Oct 2015*

- **Conducted computer vision research under NSF funding** developing machine learning algorithms for mobile robot object detection and recognition systems
- **Developed innovative "Supported_By" algorithm in MATLAB** enabling spatial relationship detection for autonomous robotics applications

Founder/Sales Director | Fatty Pocket *Norristown, PA | Sept 2011 - Jul 2013*

- **Founded and scaled digital coupon-marketing platform** for local businesses — led business development, sales teams, and strategic vendor relationships

TECHNICAL SKILLS

Forward Deployed / Solutions Engineering:
Customer Discovery • Requirements Gathering • Solution Architecture • Systems Integration • Technical Scoping & Gap Analysis • Stakeholder Management • Executive Reporting • Technical Training & AI Enablement • Bilingual (English/Korean)

Agentic / AI-Native Development:
Coding Agents (Claude Code, Codex) • Multi-Agent Orchestration • MCP (Model Context Protocol) Servers • Claude Team Administration • Prompt Engineering • Evals (RAG & Agent Evaluation) • Agent-Assisted SDLC • LangGraph Workflows

Frontend:
React.js • Next.js • TypeScript • JavaScript ES6+ • HTML5/CSS3 • Tailwind CSS • Responsive Design • i18n

Databases:
PostgreSQL (pgvector) • MySQL • DynamoDB • MongoDB • Vector Databases (Milvus) • Redis/Valkey

Enterprise Systems & Integration:
NetSuite ERP (REST API, OAuth 2.0 M2M) • Ecount ERP • Shopify (Admin API, multi-store) • Celigo iPaaS • Intercom • EDI • 3PL/WMS & Carrier APIs (FedEx, ShipStation) • Google Workspace APIs

AI/ML:
RAG Pipeline Architecture • Hybrid Search (Sparse + Dense) • Embeddings & Semantic Search • Vector Databases • Cross-encoder Reranking • LLM & Agent Evals (RAGAS, DeepEval, LLM-as-Judge, A/B & Regression) • Retrieval Metrics (NDCG, MRR, Precision/Recall@k) • Model Serving (vLLM, AWS Bedrock) • **Python:** PyTorch, Transformers, LangChain, LangGraph, LangSmith, NumPy, Pandas, scikit-learn

Backend:
Python • FastAPI/Flask • Java • Spring Boot • Node.js/Express • RESTful APIs • SSE Streaming • GraphQL • Microservices • OpenAPI/Swagger

Cloud / On-Prem / DevOps:
Google Cloud (Cloud Run, Cloud SQL, Secret Manager) • AWS (EC2, S3, Lambda, EKS, Bedrock) • Hybrid Cloud & On-Prem • OAuth 2.0 / SSO / RBAC • AI Governance & Data Privacy • Docker • Kubernetes • CI/CD • GitHub Actions • Observability

CERTIFICATIONS

Anthropic — Claude Certified Architect: Foundational (CCA-F) | Issued June 5, 2026 · Valid through June 5, 2027 · Verify: credly.com/users/john-ryu

KEY PROJECTS

TecAce AI Agent Studio - Governance-First Enterprise Agentic Platform
Hybrid search (BM25 sparse + dense vectors), LangGraph agent workflows, guardrails + multi-method evals for RAG and agents (RAGAS, DeepEval, LLM-as-judge, retrieval metrics; FinanceBench-validated), and multi-platform chat integrations — deployable on hybrid cloud or fully on-prem infrastructure for data-sensitive teams. Tech: Python, FastAPI, LangChain/LangGraph, AWS Bedrock, on-prem vLLM, Milvus

Business Compass - Bilingual Executive Operating Desk
Built for a multinational manufacturer's general manager: sales, orders, inventory, PSI planning and market intelligence with an AI agent that cites every number, daily briefs and human approval before anything goes out; connected to NetSuite, Shopify and Google Workspace. Tech: Next.js, TypeScript, PostgreSQL + pgvector, MCP, Google Cloud Run

Glaux - Personal AI Trading Desk
AI research agents, rule-based strategies, a code-only risk engine, phone approvals on Telegram and broker execution in paper mode, with an installable passkey-secured dashboard, a public demo on generated data, and a backtester that runs the same code as live. Tech: Python, PostgreSQL, Docker, Claude API

MarketScopeAI (omnidora.com) - AI Marketing Automation SaaS
AI employees that qualify leads, nurture prospects, and run marketing playbooks 24/7 for small businesses. Tech: Next.js, TypeScript, AI Agents

PlinthPrep (plinthprep.com) - CCA-F Exam Prep Platform
Practice-exam platform for Anthropic's CCA-F certification with AI-generated question variations, cheat sheets, and decision trees. Tech: Next.js, Claude API

RecompIQ (recompiq.com) - Health Data Coaching App
Body-recomposition tracker for protocols, nutrition, biomarkers, and workouts with evidence-graded insights and projections. Tech: Next.js, TypeScript, Data Visualization

EDUCATION

Bachelor of Science in Computer Science
Temple University | College of Science & Technology | Philadelphia, PA